

Proofs of the Riemann-Zeta Function

$$A \text{ Study of } \zeta(2) = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

Math 2800: Introduction to Proofs/Problem Solving

Accompanying Notes - Group Presentation

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1. Historical Background: The Basel Problem

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots$$

was first studied systematically by Pietro Mengoli in 1650, who established convergence but could not determine the value. The problem was taken up and publicised by Jakob Bernoulli in 1689, after he too failed to identify the sum. It became known as the *Basel problem*, after Basel, Switzerland, the home city of the Bernoulli mathematical dynasty.

That the series converges is not difficult to establish. A clean argument compares it to a telescoping series: since $n^2 > n(n-1)$ for all $n \geq 2$,

$$\sum_{n=2}^{\infty} \frac{1}{n^2} < \sum_{n=2}^{\infty} \frac{1}{n(n-1)} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1,$$

so the full sum is bounded above by 2. Numerical computation placed the value near $1.6449\dots$, but no closed form was apparent. The sum did not look rational, and nothing in the existing theory suggested what algebraic constant it might involve.

In 1734, Leonhard Euler announced the result:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The appearance of π in a problem with no apparent geometric content was striking. Euler's proof, detailed in Section 3, was inventive but rested on an unjustified step. Rigorous proofs came later; we present one in Section 5.

1.1. Significance of the Problem

The Basel problem is the simplest nontrivial instance of a central object in modern number theory: the Riemann zeta function $\zeta(s) = \sum n^{-s}$. The evaluation $\zeta(2) = \pi^2/6$ is both a classical result in its own right and a prototype for a family of identities connecting analysis, number theory, and the arithmetic of primes. Understanding it well is therefore a natural entry point into a broader and still-active area of mathematics.

2. The Riemann Zeta Function and $\zeta(2)$

Definition 2.1. For a real number $s > 1$, the **Riemann zeta function** is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

The restriction $s > 1$ is necessary for convergence. At $s = 1$ the series reduces to the harmonic series $\sum 1/n$, which diverges. For $s > 1$, convergence follows from the integral test: $\int_1^{\infty} x^{-s} dx = (s-1)^{-1}$.

Some representative values:

s	Exact value	Decimal approximation
2	$\pi^2/6$	1.6449...
3	no known closed form (Apéry's constant)	1.2021...
4	$\pi^4/90$	1.0823...
6	$\pi^6/945$	1.0173...

Two patterns stand out. Values at even integers involve rational multiples of powers of π ; in particular, they are transcendental. Values at odd integers beyond $s = 1$ remain mysterious. It was not until 1978 that Roger Apéry proved $\zeta(3)$ is irrational—a result so unexpected that some audience members initially disbelieved it—and whether $\zeta(3)$ (or any $\zeta(2k + 1)$ for $k \geq 2$) can be expressed in terms of π is still unknown.

The principal result of these notes is the following.

Theorem 2.2 (Euler, 1734). $\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

We present two proofs in the sections that follow.

3. Euler's Original Proof

Euler's proof proceeds by treating $\sin x$ as an infinite-degree analogue of a polynomial, factoring it over its roots, and comparing power series coefficients. We present the argument in full and identify the step that is not rigorous.

3.1. Factoring Polynomials over Their Roots

Recall that a polynomial $p(x)$ of degree n with roots $r_1, \dots, r_n$ and leading coefficient a factors as

$$p(x) = a(x - r_1)(x - r_2) \cdots (x - r_n).$$

If $p(0) \neq 0$, dividing through by $p(0) = a(-r_1) \cdots (-r_n)$ gives the normalised form

$$\frac{p(x)}{p(0)} = \prod_{k=1}^n \left(1 - \frac{x}{r_k}\right).$$

Expanding and reading off the coefficient of x yields

$$-\sum_{k=1}^n \frac{1}{r_k} = \frac{[x^1] p(x)}{p(0)},$$

i.e., the sum of reciprocal roots is determined by the first Taylor coefficient of p relative to its constant term. Euler's idea was to apply this to $\sin x$, treating it formally as a polynomial with infinitely many roots.

3.2. Applying the Factorization to $\sin x$

The Taylor series of $\sin x$ is

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Dividing by x to remove the root at 0:

$$\frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \cdots$$

This function equals 1 at $x = 0$ and has zeros at $x = \pm k\pi$ for $k = 1, 2, 3, \dots$. Grouping the conjugate pairs $(\pm k\pi)$ as difference-of-squares factors:

$$\frac{\sin x}{x} \stackrel{?}{=} \prod_{k=1}^{\infty} \left(1 - \frac{x}{k\pi}\right) \left(1 + \frac{x}{k\pi}\right) = \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right). \quad (*)$$

Euler's gap. Equation $(*)$ is the factorization step, and it is not justified by anything available to Euler in 1734. For polynomials, the factorization over roots is a theorem (the Fundamental Theorem of Algebra). For $\sin x$, an entire function defined by an infinite series, no analogous theorem existed at the time. Euler argued by analogy and verified the result numerically to high precision, but he had no proof.

The factorization is, in fact, correct, as a consequence of the *Weierstrass Factorization Theorem* (1876): every entire function can be expressed as an infinite product over its zeros, with appropriate convergence factors. For $(\sin x)/x$, whose zeros $\pm k\pi$ are simple and grow linearly, no correction factors are needed. But Euler wasn't alive for Weierstrass' greatness; in fact, the theorem beat Euler by 140 years.

3.3. Extracting $\zeta(2)$ by Coefficient Comparison

Granting $(*)$, we expand the right side as a power series and compare coefficients with the Taylor expansion on the left.

The coefficient of x^2 in $\prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right)$ is obtained by selecting the $(-x^2/k^2\pi^2)$ term from exactly one factor and 1 from all others. Summing over all k :

$$[x^2] \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right) = - \sum_{k=1}^{\infty} \frac{1}{k^2\pi^2}.$$

The Taylor series gives $[x^2] \frac{\sin x}{x} = -\frac{1}{3!} = -\frac{1}{6}$. Setting the two expressions equal:

$$- \sum_{k=1}^{\infty} \frac{1}{k^2\pi^2} = -\frac{1}{6}.$$

Multiplying both sides by $-\pi^2$:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \quad \square$$

3.4. Extracting $\zeta(4)$ by the Same Method

The coefficient of x^4 in the product $(*)$ arises from choosing the x^2 -term from exactly two factors, say the j -th and k -th:

$$[x^4] \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2 \pi^2}\right) = \sum_{1 \leq j < k < \infty} \frac{1}{j^2 k^2 \pi^4}.$$

From the Taylor series, $[x^4](\sin x)/x = 1/5! = 1/120$. Using the algebraic identity

$$\left(\sum_{n=1}^{\infty} \frac{1}{n^2}\right)^2 = \sum_{n=1}^{\infty} \frac{1}{n^4} + 2 \sum_{1 \leq j < k} \frac{1}{j^2 k^2}$$

together with $\sum 1/n^2 = \pi^2/6$ already established, one obtains

$$\zeta(4) = \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

4. Discussion of Euler's Proof

Several observations are worth making before moving on.

Euler's epistemic position. Euler knew the factorization step was an assumption. His justification was indirect: the result agreed with extensive numerical computation (he had computed $\sum 1/n^2$ to many decimal places and recognised it as $\pi^2/6$), and the same method yielded further values $\zeta(4), \zeta(6), \dots$ that also agreed numerically. In the eighteenth century, such convergence of evidence was commonly regarded as constituting a proof, or at least sufficient grounds to state a theorem. The modern demand for deductive completeness was not yet standard.

Even granting the product formula $(*)$, extracting the x^2 -coefficient by formal term-by-term manipulation requires that the infinite product converges uniformly on compact sets as a power series—a condition that again requires verification. This too follows from the Weierstrass theory: the product converges uniformly on compact subsets of $\mathbb{C}$, and so can be manipulated term-by-term.

Lasting influence. Despite the gap, Euler's proof introduced a style of reasoning—linking analytic functions to their zeros via infinite products—that became foundational. The *Euler product* for the Riemann zeta function,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}, \quad s > 1,$$

is a rigorous theorem following from unique prime factorization, and stands as the most important example of the same idea. It is the bridge between analysis and the distribution of primes.

5. A Rigorous Proof via Fourier Series

We now prove Theorem 2.2 rigorously. The argument uses only real analysis: integration, uniform convergence, and a basic theorem about trigonometric series.

5.1. Fourier Series

Let $f : [-\pi, \pi] \rightarrow \mathbb{R}$ be piecewise continuous. The **Fourier series** of f is the formal expansion

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)),$$

where the **Fourier coefficients** of f are

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx, \quad n \geq 0,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx, \quad n \geq 1.$$

These formulas arise from the *orthogonality* of the trigonometric system.

Lemma 5.1 (Orthogonality of Sine and Cosine). *For integers $m, n \geq 1$,*

$$\int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx = \pi \delta_{mn}, \quad (1)$$

$$\int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx = \pi \delta_{mn}, \quad (2)$$

$$\int_{-\pi}^{\pi} \cos(mx) \sin(nx) dx = 0, \quad (3)$$

where $\delta_{mn} = 1$ if $m = n$ and 0 otherwise. For $n \geq 1$,

$$\int_{-\pi}^{\pi} \cos(nx) dx = \int_{-\pi}^{\pi} \sin(nx) dx = 0. \quad (4)$$

To see why: the identity $2 \cos(mx) \cos(nx) = \cos((m-n)x) + \cos((m+n)x)$ reduces each integral to $\int_{-\pi}^{\pi} \cos(kx) dx$, which equals 2π if $k = 0$ and 0 if $k \neq 0$. The formula for b_n is then obtained by multiplying both sides of the Fourier expansion by $\sin(nx)$, integrating over $[-\pi, \pi]$, and using orthogonality: every term on the right vanishes except $b_n \pi$.

Convergence of the Fourier series to f is guaranteed for nice functions by the following theorem, which we state without proof.

Theorem 5.2 (Dirichlet, 1829). *If $f : [-\pi, \pi] \rightarrow \mathbb{R}$ is piecewise smooth (continuous except at finitely many points, with one-sided derivatives everywhere), then for all $x \in (-\pi, \pi)$,*

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx)) = \frac{f(x^+) + f(x^-)}{2}.$$

In particular, the series converges to $f(x)$ at every point of continuity.

5.2. Parseval's Theorem

The key tool in our proof is the following identity.

Theorem 5.3 (Parseval, 1799). *If $f : [-\pi, \pi] \rightarrow \mathbb{R}$ is piecewise continuous with Fourier coefficients $\{a_n\}$ and $\{b_n\}$, then*

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2).$$

Remark. Parseval's theorem is an infinite-dimensional analogue of the Pythagorean theorem. Think of f as a vector in the space of square-integrable functions, with $\{\frac{1}{\sqrt{2}}, \cos(nx), \sin(nx)\}$ as an orthonormal basis (after normalising by $1/\sqrt{\pi}$). The left side is the squared norm $\|f\|^2$, and the right side is the sum of squared coordinates. The theorem says the basis is complete: no “energy” is lost in the expansion. The proof follows from uniform L^2 convergence of the partial sums, a standard result in real analysis.

5.3. Proof of $\zeta(2) = \pi^2/6$

Before beginning the formal proof, it is worth mentioning to explain *why* the function $f(x) = x$ is chosen. This is the key design decision in the proof.

We need a function f on $[-\pi, \pi]$ satisfying three criteria simultaneously:

- (i) Its Fourier coefficients involve $1/n$, so that their squares give $1/n^2$ — and summing over n produces $\zeta(2)$.
- (ii) The integral $\frac{1}{\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx$ can be evaluated in closed form by elementary calculus.
- (iii) The function is *odd*, meaning $f(-x) = -f(x)$, so that all cosine coefficients a_n vanish and the Parseval sum reduces to $\sum b_n^2$ alone.

The function $f(x) = x$ satisfies all three. Its sine coefficients will be $b_n = 2(-1)^{n+1}/n$, giving $b_n^2 = 4/n^2$, and $\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = 2\pi^2/3$ is a routine polynomial integral.

Proof overview. Define $f(x) = x$ on $(-\pi, \pi)$. Because f is an odd function, all cosine Fourier coefficients vanish: $a_n = 0$ for all $n \geq 0$. A computation using integration by parts yields the sine coefficients $b_n = 2(-1)^{n+1}/n$. Parseval's Theorem 5.3 then gives

$$\underbrace{\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx}_{= 2\pi^2/3} = \underbrace{\sum_{n=1}^{\infty} b_n^2}_{= 4\zeta(2)}.$$

Solving immediately: $\zeta(2) = \pi^2/6$.

The entire argument is a single chain:

choose f $\longrightarrow$ compute b_n $\longrightarrow$ apply Parseval $\longrightarrow$ solve for $\zeta(2)$.

Proof of Theorem 2.2. Let $f(x) = x$ on $(-\pi, \pi)$.

Step 1: Compute the Fourier coefficients.

Since f is odd, all cosine coefficients vanish: $a_n = 0$ for all $n \geq 0$. For the sine coefficients, integrate by parts with $u = x$, $dv = \sin(nx) dx$:

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx \\ &= \frac{1}{\pi} \left[-\frac{x \cos(nx)}{n} \Big|_{-\pi}^{\pi} + \frac{1}{n} \int_{-\pi}^{\pi} \cos(nx) dx \right]. \end{aligned}$$

The boundary term evaluates to $-\frac{1}{n} [\pi \cos(n\pi) + \pi \cos(n\pi)] = -\frac{2\pi \cos(n\pi)}{n}$, and the remaining integral $\int_{-\pi}^{\pi} \cos(nx) dx = 0$ for $n \geq 1$. Using $\cos(n\pi) = (-1)^n$:

$$b_n = \frac{1}{\pi} \cdot \frac{-2\pi(-1)^n}{n} = \frac{2(-1)^{n+1}}{n}.$$

Step 2: Confirm pointwise convergence.

The function $f(x) = x$ is smooth on $(-\pi, \pi)$, so Theorem 5.2 gives, for all $x \in (-\pi, \pi)$:

$$x = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin(nx).$$

Step 3: Apply Parseval's theorem.

With $a_n = 0$ for all n , Theorem 5.3 gives:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \sum_{n=1}^{\infty} b_n^2.$$

Left side:

$$\frac{1}{\pi} \int_{-\pi}^{\pi} x^2 dx = \frac{1}{\pi} \cdot \frac{2\pi^3}{3} = \frac{2\pi^2}{3}.$$

Right side:

$$\sum_{n=1}^{\infty} b_n^2 = \sum_{n=1}^{\infty} \frac{4}{n^2} = 4\zeta(2).$$

Therefore $\frac{2\pi^2}{3} = 4\zeta(2)$, giving $\zeta(2) = \frac{\pi^2}{6}$. □

6. Discussion of the Fourier Proof

The Fourier proof uses three components, which we focused on prior: the orthogonality of the trigonometric system (a direct computation), Dirichlet's convergence theorem for piecewise smooth functions, and Parseval's theorem. The computation of b_n requires only integration by parts. No unjustified interchange of limits occurs.

Pointwise convergence is not strictly needed. The proof uses Parseval's theorem, which holds for piecewise continuous functions and does not require pointwise convergence of the Fourier series. Step 2 (Dirichlet's theorem) is included for completeness and to confirm that the Fourier expansion we derived is the correct one, but the proof of $\zeta(2) = \pi^2/6$ uses only Parseval.

7. Analytic Continuation and Special Values of ζ

Definition 2.1 restricts the zeta function to $s > 1$. It is natural to ask whether the *function* can be extended beyond this range. The answer is yes, via a process from complex analysis.

A function defined on an open region of the complex plane and complex-differentiable at every point is called **holomorphic**. Holomorphic functions are extraordinarily rigid: if two holomorphic functions agree on any open set, they agree everywhere their domains overlap. It follows that if there is a holomorphic function on a larger domain that agrees with $\zeta(s)$ on $\{\operatorname{Re}(s) > 1\}$, then it is unique. Such an extension is called an **analytic continuation**.

Using the functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) \zeta(1-s),$$

where Γ is the Gamma function (the unique smooth extension of $n! = n \cdot (n-1)!$ to positive real numbers, satisfying $\Gamma(n) = (n-1)!$), one can extend ζ to a meromorphic function on all of $\mathbb{C}$ with a single simple pole at $s = 1$. This extended function is what number theorists mean by “the Riemann zeta function.”

The special values of the extended function at non-positive integers are expressed in terms of the **Bernoulli numbers** B_n , defined by the generating function

$$\frac{t}{e^t - 1} = \sum_{n=0}^{\infty} B_n \frac{t^n}{n!}.$$

The first few values are $B_0 = 1$, $B_1 = -1/2$, $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, with $B_{2k+1} = 0$ for $k \geq 1$. The relevant formula is:

$$\zeta(-n) = -\frac{B_{n+1}}{n+1}, \quad n = 0, 1, 2, \dots$$

giving, for instance, $\zeta(0) = -1/2$ and $\zeta(-1) = -1/12$. These are values of the analytically continued function, not sums of divergent series.

8. Further Consequences and Generalizations

8.1. Values at Even Positive Integers

The method of Section 5 extends: applying Parseval to $f(x) = x^k$ (or working systematically through the Laurent expansion of $\pi \cot(\pi z)$) yields closed forms for all $\zeta(2k)$.

Theorem 8.1 (Euler). *For every positive integer k ,*

$$\zeta(2k) = \frac{(-1)^{k+1} (2\pi)^{2k} B_{2k}}{2(2k)!}.$$

k	B_{2k}	$\zeta(2k)$	Decimal
1	1/6	$\pi^2/6$	1.6449...
2	-1/30	$\pi^4/90$	1.0823...
3	1/42	$\pi^6/945$	1.0173...
4	-1/30	$\pi^8/9450$	1.0041...
5	5/66	$\pi^{10}/93555$	1.0010...

Every $\zeta(2k)$ is a rational multiple of π^{2k} , hence transcendental. The denominators of these fractions are related to Bernoulli numbers, which appear throughout number theory: in formulae for class numbers of number fields, in the Kummer criterion for regular primes, and in the study of Fermat's Last Theorem.

8.2. The Alternating Sum: The Dirichlet Eta Function

Alternating the signs of the summands gives a related but more broadly convergent series.

Definition 8.2. The **Dirichlet eta function** is

$$\eta(s) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s} = 1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \cdots$$

By the alternating series test, $\eta(s)$ converges for all $s > 0$, extending convergence one unit to the left of ζ .

Proposition 8.3. For $s > 1$, $\eta(s) = (1 - 2^{1-s}) \zeta(s)$.

Proof. Starting from $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, group even and odd terms:

$$\zeta(s) = \sum_{\text{odd } n} \frac{1}{n^s} + \sum_{n=1}^{\infty} \frac{1}{(2n)^s}.$$

The even-term sum equals $2^{-s}\zeta(s)$. Therefore $\sum_{\text{odd}} \frac{1}{n^s} = (1 - 2^{-s})\zeta(s)$. Adding odd terms with + sign and even terms with - sign:

$$\eta(s) = \sum_{\text{odd}} \frac{1}{n^s} - \sum_{n=1}^{\infty} \frac{1}{(2n)^s} = (1 - 2^{-s})\zeta(s) - 2^{-s}\zeta(s) = (1 - 2^{1-s})\zeta(s). \quad \square$$

□

Rearranging, $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ for $s \neq 1$. At $s = 1$, the numerator $\eta(1) = \ln 2$ (the alternating harmonic series) and the denominator $(1 - 2^{1-s}) \rightarrow 0$, reflecting the simple pole of ζ .

Selected values:

$$\begin{aligned} \eta(1) &= \ln 2, \\ \eta(2) &= \left(1 - \frac{1}{2}\right) \zeta(2) = \frac{\pi^2}{12}, \\ \eta(4) &= \left(1 - \frac{1}{8}\right) \zeta(4) = \frac{7\pi^4}{720}. \end{aligned}$$

8.3. Dirichlet L -Functions

The Riemann zeta function is a member of a richer family of Dirichlet series.

Definition 8.4. A **Dirichlet character mod q** is a function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ that is:

- periodic with period q : $\chi(n + q) = \chi(n)$;
- completely multiplicative: $\chi(mn) = \chi(m)\chi(n)$;
- zero when $\gcd(n, q) > 1$, and nonzero otherwise.

The associated **Dirichlet L -function** is

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \quad s > 1.$$

When $q = 1$ and $\chi = \chi_0$ is the trivial character ($\chi_0(n) = 1$ for all n), we recover $L(s, \chi_0) = \zeta(s)$.

As a concrete nontrivial example, the unique non-principal character mod 4 is:

$$\chi_4(n) = \begin{cases} 0 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \equiv 1 \pmod{4}, \\ -1 & \text{if } n \equiv 3 \pmod{4}. \end{cases}$$

This gives

$$L(1, \chi_4) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4},$$

which is Leibniz's formula for $\pi/4$, recognisable as a special value of an L -function. It can be proved via the Fourier series method applied to $f(x) = \pi/4 - x/2$ on $(0, \pi)$, or directly from the antiderivative of $1/(1 + t^2)$.

Like $\zeta(s)$, each $L(s, \chi)$ has an Euler product:

$$L(s, \chi) = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p)p^{-s}},$$

which converges for $s > 1$ and reflects the unique factorisation of the integers together with the multiplicativity of χ .

8.4. The Riemann Hypothesis

The study of $\zeta(s)$ as a complex function leads to one of the most famous open problems in mathematics. Riemann observed in 1859 that the distribution of prime numbers is governed in a precise way by the zeros of the analytically continued $\zeta(s)$ in the complex plane. Specifically, letting $\pi(x)$ denote the number of primes up to x , there is an explicit formula

$$\pi(x) = \text{Li}(x) - \sum_{\rho: \zeta(\rho)=0} \text{Li}(x^\rho) + \cdots$$

where $\text{Li}(x) = \int_2^x dt/\ln t$ and the sum runs over the nontrivial zeros of ζ . The zeros of ζ in the *critical strip* $\{0 < \text{Re}(s) < 1\}$ are all conjectured to lie on the *critical line* $\text{Re}(s) = 1/2$.

Theorem 8.5 (Riemann Hypothesis, conjectured 1859). *All nontrivial zeros of $\zeta(s)$ satisfy $\operatorname{Re}(s) = 1/2$.*

This remains unproved. It is a Millennium Prize Problem (Clay Mathematics Institute, prize: one million USD or some huge sum idk). As of 2024, it has been numerically verified for the first 10^{13} zeros, all on the critical line.

References.

- J. Stillwell, *Mathematics and Its History* (3rd ed., Springer). Thorough historical treatment of Euler's work and the Basel problem.
- E. Stein and R. Shakarchi, *Fourier Analysis: An Introduction* (Princeton Lectures in Analysis, Vol. 1). Rigorous development of Fourier series, including Parseval and Dirichlet's convergence theorem.
- T. Apostol, *Introduction to Analytic Number Theory* (Springer). Complete proofs of the Euler product, Dirichlet's theorem, and L -function special values.
- H. M. Edwards, *Riemann's Zeta Function* (Dover). Definitive reference for the analytic theory of $\zeta(s)$ and the Riemann Hypothesis.